

LAB

Topic 7: Trigger



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Trigger

Triggers are used to enforce business rules, maintain data integrity, and perform additional actions based on changes to the data.

Given a business rule BR#1, the following steps enable determining and implementing triggers to ensure compliance with the business rule:

- related to the business rule.
- Analyzing the risk associated with the business rule, specifically identifying the primitive actions such as insert, delete, and update on the tables within the scope that could potentially violate the business rule.

The scope and risks of a business rule can be described in the following table. (+) denotes that the corresponding action may violate the business rule. (-) denotes that this action does not affect the business rule.

Tables within the scope	Insert	Delete	Update
A	+	-	+ (attr1, attr2)
B	-	+	-

Suppose that the above table describes the scope and risks of the business rule BR#1. This business rule may be violated when:

- (1) adding a new row into table A and updating the two columns attr1 & attr2 of table A;
- (2) deleting a row from table B.

Two (or three) different triggers will be implemented on tables A and B to ensure compliance with business rule BR#1.

```
CREATE TRIGGER trigger_name
ON table_name -- Specifying the name of the table to which the trigger belongs.
FOR INSERT|DELETE|UPDATE -- Specifying actions that can invoke the trigger.
AS
BEGIN
/*Trigger logic here
--Note: Whenever a trigger belonging to table A is invoked, two temporary
(virtual) tables with the same structure as A are available:
- INSERTED: contains the new rows being inserted or updated
- DELETED: contains the old rows being deleted or updated */

/* ERROR MESSAGE
```

```
--Using RAISERROR
RAISERROR (msg, severity, state)
-- msg: A custom error message (string) or a message ID from sys.messages
-- severity: Indicates the seriousness of the error (range: 0-25). The severity
level of an error can range from 0-25.
-- state: A user-defined code (0-255) to indicate where the error occurred. It
can range from 0-255. */

END
```

Recommended implementation process

Write the business rule in a precise and testable form.

1. Identify the complete scope and risk table.
2. Choose the table and events for each required trigger.
3. Write the violation query first as a normal SELECT statement.
4. Place the violation query inside IF EXISTS.
5. Test one valid row, one invalid row, and a multi-row statement.

Example: Identifying the scope, risks and implementing triggers to ensure this business rule

“A department can have a maximum of 10 instructors.”

Tables within the scope	Insert	Delete	Update
Instructor	+	-	+ (department_id)
Department	-	-	-

```
CREATE TRIGGER trg_dept_id ON Instructor
FOR INSERT, UPDATE
AS
BEGIN
    IF EXISTS(SELECT s.department_id FROM INSTRUCTOR s
    Group by s.department_id
    Having count(s.instructor_id)>10)
```

```
BEGIN
    RAISERROR ('Error', 16, 1)
    ROLLBACK;
END
END
```